

Type "HXXX" Faults

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
H001: GS30, GS32, GS46 & GS47	H001:COILFAULT PLAT UP1:BAT-	Short circuit of the platform up #1 circuit to battery negative.	- Short circuit in platform up #1 harness. - Platform up #1 coil short circuit. - GCON ECM.	Platform up function inhibited.
H002: GS30, GS32, GS46 & GS47	H002:COILFAULT PLAT UP1:OPEN	Open circuit in the platform up #1 circuit.	- Short circuit in platform up #1 harness. - Platform up #1 coil open circuit. - GCON ECM.	Platform up function inhibited.
H003: GS30, GS32, GS46 & GS47	H003:COILFAULT PLAT UP1:BAT+	Short circuit of the platform up #1 circuit to battery positive.	- Short circuit in platform up #1 harness. - Platform up #1 coil short circuit. - GCON ECM.	All functions inhibited except platform down.
H009: GS30, GS32, GS46 & GS47	H009:COILFAULT PLAT DOWN1:BAT+	Short circuit of the platform down #1 circuit to battery positive.	- Short circuit in platform down #1 harness. - Platform down #1 coil short circuit. - GCON ECM.	All functions inhibited.
H013: GS30, GS32, GS46 & GS47	H013:COILFAULT DRIVE FWD1:BAT-	Short circuit of the drive forward #1 circuit to battery negative.	- Short circuit in drive forward #1 harness. - Drive forward #1 coil short circuit. - GCON ECM.	Drive forward function inhibited.
H014: GS30, GS32, GS46 & GS47	H014:COILFAULT DRIVE FWD1:OPEN	Open circuit in the drive forward #1 circuit.	- Open circuit in drive forward #1 harness. - Drive forward #1 coil open circuit. - GCON ECM.	Drive forward function inhibited.
H015: GS30, GS32, GS46 & GS47	H015:COILFAULT DRIVE FWD1:BAT+	Short circuit of the drive forward #1 circuit to battery positive.	- Short circuit in drive forward #1 harness. - Drive forward #1 coil short circuit. - GCON ECM.	All functions inhibited except platform down.
H019: GS30, GS32, GS46 & GS47	H019:COILFAULT DRIVE REV1:BAT-	Short circuit of the drive reverse #1 circuit to battery negative.	- Short circuit in drive reverse #1 harness. - Drive reverse #1 coil short circuit. - GCON ECM.	Drive reverse function inhibited.
H020: GS30, GS32, GS46 & GS47	H020:COILFAULT DRIVE REV1:OPEN	Open circuit in the drive reverse #1 circuit.	- Open circuit in drive reverse #1 harness. - Drive reverse #1 coil open circuit. - GCON ECM.	Drive reverse function inhibited.
H021: GS30, GS32, GS46 & GS47	H021:COILFAULT DRIVE REV1:BAT+	Short circuit of the drive reverse #1 circuit to battery positive.	- Short circuit in drive reverse #1 harness. - Drive reverse #1 coil short circuit. - GCON ECM.	All functions inhibited except platform down.
H027: GS30, GS32, GS46 & GS47	H027:COILFAULT DRIVE STEER RIGHT:BAT+	Short circuit of the steer right circuit to battery positive.	- Short circuit in steer right harness. - Steer right coil short circuit. - GCON ECM.	All functions inhibited except platform down.
H030: GS30, GS32, GS46 & GS47	H030:COILFAULT DRIVE STEER LEFT:BAT+	Short circuit of the steer left circuit to battery positive.	- Short circuit in steer left harness. - Steer left coil short circuit. - GCON ECM.	All functions inhibited except platform down.
H033: GS30, GS32, GS46 & GS47	H033:COILFAULT HI/LO SPEED:BAT+	Short circuit of the hi/lo speed coil to battery positive.	- Short circuit in Hi/Lo speed coil harness. - Hi/Lo speed coil short circuit. - GCON ECM.	Hi/Lo speed coil disabled. Machine will operate in low speed mode.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
H049: GS-3232	H049:COILFAULT O/R EXTEND:BAT-	Short circuit of the outrigger extend coil to battery negative.	• Short circuit in outrigger extend coil harness. • Outrigger extend coil short circuit. • GCON ECM.	Only outrigger extend function disabled.
H050: GS-3232	H050:COILFAULT O/R EXTEND:OPEN	Open circuit in the outrigger extend coil circuit.	• Short circuit in outrigger extend coil harness. • Outrigger extend coil open circuit. • GCON ECM.	Only outrigger extend function disabled.
H051: GS-3232	H051:COILFAULT O/R EXTEND:BAT+	Short circuit of the outrigger extend coil to battery positive.	• Short circuit in outrigger extend coil harness. • Outrigger extend coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H052: GS-3232	H052:COILFAULT O/R RETRACT:BAT-	Short circuit of the outrigger retract coil to battery negative.	• Short circuit in outrigger retract coil harness. • Outrigger retract coil short circuit. • GCON ECM.	Only outrigger retract function disabled.
H053: GS-3232	H053:COILFAULT O/R RETRACT:OPEN	Open circuit in the outrigger retract coil circuit.	• Short circuit in outrigger retract coil harness. • Outrigger retract coil open circuit. • GCON ECM.	Only outrigger retract function disabled.
H054: GS-3232	H054:COILFAULT O/R RETRACT:BAT+	Short circuit of the outrigger retract coil to battery positive.	• Short circuit in outrigger retract coil harness. • Outrigger retract coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H057: GS-3232	H057:COILFAULT LF RIGGER:BAT+	Short circuit of the left front outrigger coil to battery positive.	• Short circuit in left front outrigger coil harness. • Left front outrigger coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H060: GS-3232	H060:COILFAULT LR RIGGER:BAT+	Short circuit of the left rear outrigger coil to battery positive.	• Short circuit in left rear outrigger coil harness. • Left rear outrigger coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H063: GS-3232	H063:COILFAULT RF RIGGER:BAT+	Short circuit of the right front outrigger coil to battery positive.	• Short circuit in right front outrigger coil harness. • Right front outrigger coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H066: GS-3232	H066:COILFAULT RR RIGGER:BAT+	Short circuit of the right rear outrigger coil to battery positive.	• Short circuit in right rear outrigger coil harness. • Right rear outrigger coil short circuit. • GCON ECM.	All functions inhibited except platform down.
H067: GS30, GS32, GS46 & GS47	H067:FAULT MC ENABLE:BAT-	Short circuit of the motor controller circuit to battery negative.	• Short circuit in motor controller enable wire. • Motor Controller. • Contactor coil shorted to battery negative. • GCON ECM.	All functions inhibited except platform down.
H069: GS30, GS32, GS46 & GS47	H069:FAULT MC ENABLE:BAT+	Short circuit of the motor controller circuit to battery positive.	• Short circuit in motor controller enable wire. • Motor Controller. • Contactor Coil. • GCON ECM.	All functions inhibited except platform down.
H070: GS30, GS32, GS46 & GS47	H070:FAULT MC THROTTLE:BAT-	Short circuit of the motor controller throttle circuit to battery negative.	• Short circuit in motor controller throttle wire. • Motor Controller. • GCON ECM.	All functions inhibited except platform down.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
H072: GS30, GS32, GS46 & GS47	H072:FAULT MC THROTTLE:BAT+	Short circuit of the motor controller throttle circuit to battery positive.	• Short circuit in motor controller throttle wire. • Motor Controller. • GCON ECM.	All functions inhibited except platform down.
H074: GS-3232	H074:COILFAULT LF RIGGER	Short circuit of the left front outrigger circuit to battery positive/negative or open circuit.	• Short or open circuit in left front outrigger harness. • Left front outrigger coil short or open circuit. • GCON ECM.	Left front outrigger function inhibited.
H075: GS-3232	H075:COILFAULT LR RIGGER	Short circuit of the left rear outrigger circuit to battery positive/negative or open circuit.	• Short or open circuit in left rear outrigger harness. • Left rear outrigger coil short or open circuit. • GCON ECM.	Left rear outrigger function inhibited.
H076: GS-3232	H076:COILFAULT RF RIGGER	Short circuit of the right front outrigger circuit to battery positive/negative or open circuit.	• Short or open circuit in right front outrigger harness. • Right front outrigger coil short or open circuit. • GCON ECM.	Right front outrigger function inhibited.
H077: GS-3232	H077:COILFAULT RR RIGGER	Short circuit of the right rear outrigger circuit to battery positive/negative or open circuit.	• Short or open circuit in right rear outrigger harness. • Right rear outrigger coil short or open circuit. • GCON ECM.	Right rear outrigger function inhibited.
H078: GS30, GS32, GS46 & GS47	H078:COILFAULT PLAT DOWN1	Short circuit of the platform down #1 circuit to battery positive/negative or open circuit.	• Short or open circuit in platform down #1 harness. • Platform down #1 coil short or open circuit. • GCON ECM.	Platform down function inhibited.
H079: GS32, GS46 & GS47	H079:COILFAULT HI/LO SPEED	Short circuit of the HI/LO speed circuit to battery positive/negative or open circuit.	• Short circuit in HI/LO speed harness. • HI/LO speed coil short or open circuit. • GCON ECM.	Hi/Lo speed coil disabled. Machine will operate in high speed mode if open circuit or short with battery negative. Machine will operate in low speed mode if short with battery positive.
H080: GS30, GS32, GS46 & GS47	H080:COILFAULT STEER LEFT	Short circuit of the steer left circuit to battery negative or open circuit.	• Short or open circuit in steer left harness. • Steer left coil short or open circuit. • GCON ECM.	Steer left function inhibited.
H081: GS30, GS32, GS46 & GS47	H081:COILFAULT STEER RIGHT	Short circuit of the steer right circuit to battery negative or open circuit.	• Short or open circuit in steer right harness. • Steer right coil short or open circuit. • GCON ECM.	Steer right function inhibited.
H102: GS-4047	H102:COILFAULT RELIEF VALVE:BAT+	Short circuit of the relief valve circuit to battery positive.	• Short circuit in relief valve harness. • Relief valve coil short circuit. • GCON ECM.	All functions inhibited except platform down while above limit switch.
H103: GS-4047	H103:COILFAULT RELIEF VALVE:BAT-	Short circuit of the relief valve circuit to battery negative.	• Short circuit in relief valve harness. • Relief valve coil short circuit. • GCON ECM.	All functions inhibited except platform down while above limit switch.
H104: GS-4047	H104:COILFAULT RELIEF VALVE:OPEN	Relief valve circuit open.	• Open circuit in relief valve harness. • Relief valve coil open circuit. • GCON ECM.	All functions inhibited except platform down while above limit switch.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
H106: GS-2646AV	H106:COILFAULT DECK EXTEND:BAT-	Short circuit of the platform extend circuit to battery negative.	• Short circuit in PED harness. • PED extend valve coil short circuit. • GCON ECM.	All functions inhibited.
H107: GS-2646AV	H107:COILFAULT DECK EXTEND:OPEN	Open circuit in the platform extend circuit.	• Open circuit in PED harness. • PED extend valve coil open circuit. • GCON ECM.	All functions inhibited.
H108: GS-2646AV	H108:COILFAULT DECK EXTEND:BAT+	Short circuit of the platform extend circuit to battery positive.	• Short circuit in PED harness. • PED extend valve coil short circuit. • GCON ECM.	All functions inhibited.
H109: GS-2646AV	H109:COILFAULT DECK RETRACT:BAT-	Short circuit of the platform retract circuit to battery negative.	• Short circuit in PED harness. • PED retract valve coil short circuit. • GCON ECM.	All functions inhibited.
H110: GS-2646AV	H110:COILFAULT DECK RETRACT:OPEN	Open circuit in the platform retract circuit.	• Open circuit in PED harness. • PED retract valve coil open circuit. • GCON ECM.	All functions inhibited.
H111: GS-2646AV	H111:COILFAULT DECK RETRACT:BAT+	Short circuit of the platform retract circuit to battery positive.	• Short circuit in PED harness. • PED retract valve coil short circuit. • GCON ECM.	All functions inhibited.
H114: GS-2646AV	H114:COILFAULT DECK ENABLE:BAT+	Short circuit of the platform enable circuit to battery positive.	• Short circuit in PED harness. • PED enable valve coil short circuit. • GCON ECM.	All functions inhibited.
H115: GS-2646AV	H115:COILFAULT DECK ENABLE	Short circuit of the platform enable circuit.	• Short circuit in PED harness. • GCON ECM.	All functions inhibited.

Type "PXXX" Faults

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
P001: GS30, GS32, GS46 & GS47	P001:PWR FAULT SW PWR1:BAT-	Short circuit of the switched power #1 circuit to battery negative.	• Short circuit in switched power #1, down limit switch, pothole limit switch, digital tilt switch harness. • GCON ECM.	All functions inhibited.
P003: GS30, GS32, GS46 & GS47	P003:PWR FAULT SW PWR1:BAT+	Short circuit of the switched power #1 circuit to battery positive.	• Short circuit in switched power #1, down limit switch, pothole limit switch, digital tilt switch harness. • GCON ECM.	All functions inhibited.
P004: GS30, GS32, GS46 & GS47	P004:DEVICEFAULT T HORN:BAT-	Short circuit of the automotive horn circuit to battery negative.	• Short circuit in automotive horn harness. • Automotive horn short circuit. • GCON ECM.	Automotive horn inhibited.
P005: GS30, GS32, GS46 & GS47	P005:DEVICEFAULT T HORN:OPEN	Open circuit of the automotive horn circuit.	• Open circuit in automotive horn harness. • Automotive horn open circuit. • GCON ECM.	Automotive horn inhibited.
P006: GS30, GS32, GS46 & GS47	P006:DEVICEFAULT T HORN:BAT+	Short circuit of the automotive horn circuit to battery positive.	• Short circuit in automotive horn harness. • Automotive horn short circuit. • GCON ECM.	Automotive horn inhibited.
P007: GS30, GS32, GS46 & GS47	P007:DEVICEFAULT T GCON ALARM:BAT-	Short circuit of the GCON alarm circuit to battery negative.	• Short circuit in GCON alarm harness. • GCON alarm short circuit. • GCON ECM.	GCON alarm inhibited.
P009: GS30, GS32, GS46 & GS47	P009:DEVICEFAULT T GCON ALARM:BAT+	Short circuit of the GCON alarm circuit to battery positive.	• Short circuit in GCON alarm harness. • GCON alarm short circuit. • GCON ECM.	GCON alarm inhibited.
P013: GS30, GS32, GS46 & GS47	P013:PWR FAULT PCON PWRET:BAT-	Short circuit of the PCON power return circuit to battery negative.	• Short circuit in PCON power return harness. • GCON ECM.	All functions inhibited.
P015: GS30, GS32, GS46 & GS47	P015:PWR FAULT PCON PWRET:BAT+	Short circuit of the PCON power return circuit to battery positive.	• Short circuit in PCON power return harness. • GCON ECM.	All functions inhibited.
P018: GS30, GS32, GS46 & GS47	018:PWR FAULT PCON POWER:BAT-	Short circuit of the PCON power circuit to battery negative.	• Short circuit in PCON power harness. • GCON ECM.	All functions inhibited.
P019: GS30, GS32, GS46 & GS47	018:PWR FAULT PCON POWER:BAT+	Short circuit of the PCON power circuit to battery positive.	• Short circuit in PCON power harness. • GCON ECM.	All functions inhibited.

Type "UXXX" Faults

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
U001: GS30, GS32, GS46 & GS47	U001:SWITCHFAULT GCON MAIN FTN EN	Short circuit of the GCON main function enable switch at system startup.	• Short circuit of the GCON main function enable switch. • GCON ECM.	All GCON functions inhibited.
U002: GS30, GS32, GS46 & GS47	U002:SWITCHFAULT GCON PLAT UP	Short circuit of the GCON up directional switch at system startup.	• Short circuit of the GCON up directional switch. • GCON ECM.	All GCON functions inhibited except platform down.
U003: GS30, GS32, GS46 & GS47	U003:SWITCHFAULT GCON PLAT DOWN	Short circuit of the GCON down directional switch at system startup.	• Short circuit of the GCON down directional switch. • GCON ECM.	All GCON functions inhibited except platform up.
U004: GS30, GS32, GS46 & GS47	U004:SWITCHFAULT GCON LCD UP	Short circuit of the GCON LCD scroll up switch at system startup.	• Short circuit of the GCON LCD scroll up switch. • GCON ECM.	All GCON LCD menu functions inhibited.
U005: GS30, GS32, GS46 & GS47	U005:SWITCHFAULT GCON LCD DOWN	Short circuit of the GCON LCD scroll down switch at system startup.	• Short circuit of the GCON LCD scroll down switch. • GCON ECM.	All GCON LCD menu functions inhibited.
U006: GS30, GS32, GS46 & GS47	U006:SWITCHFAULT GCON LCD ENTER	Short circuit of the GCON LCD enter switch at system startup.	• Short circuit of the GCON LCD enter switch. • GCON ECM.	All GCON LCD menu functions inhibited.
U007: GS30, GS32, GS46 & GS47	U007:SWITCHFAULT GCON LCD ESCAPE	Short circuit of the GCON LCD escape switch at system startup.	• Short circuit of the GCON LCD escape switch. • GCON ECM.	All GCON LCD menu functions inhibited.
U014: GS30, GS32, GS46 & GS47	U014:SWITCHFAULT PCON DRIVE EN	Short circuit of the PCON drive enable switch at system startup.	• Short circuit of the PCON drive enable switch. • GCON ECM.	All PCON drive and steer functions inhibited.
U015: GS30, GS32, GS46 & GS47	U015:SWITCHFAULT PCON STEER LEFT	Short circuit of the PCON steer left switch at system startup.	• Short circuit of the PCON steer left switch. • GCON ECM.	All PCON drive and steer functions inhibited.
U016: GS30, GS32, GS46 & GS47	U016:SWITCHFAULT PCON STEER RIGHT	Short circuit of the PCON steer right switch at system startup.	• Short circuit of the PCON steer right switch. • GCON ECM.	All PCON drive and steer functions inhibited.
U017: GS30, GS32, GS46 & GS47	U017:SWITCHFAULT PCON HORN	Short circuit of the PCON horn switch at system startup.	• Short circuit of the PCON horn switch. • GCON ECM.	PCON horn switch function inhibited.
U018: GS30, GS32, GS46 & GS47	U018:SWITCHFAULT PCON LO DRIV SPD	Short circuit of the PCON low drive speed switch at system startup.	• Short circuit of the PCON low drive speed switch. • GCON ECM.	The machine is limited to low drive speed.
U019: GS30, GS32, GS46 & GS47	U019:SWITCHFAULT PCON LO LIFT SPD	Short circuit of the PCON low lift speed switch at system startup.	• Short circuit of the PCON low lift speed switch. • GCON ECM.	PCON platform up and down functions inhibited.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
U020: GS30, GS32, GS46 & GS47	U020:SWITCHFAULT PCON HI LIFT SPD	Short circuit of the PCON high lift speed switch at system startup.	• Short circuit of the PCON hi lift speed switch. • GCON ECM.	PCON platform up and down functions inhibited.
U021: GS30, GS32, GS46 & GS47	U021:SWITCHFAULT PCON UP	Short circuit of the PCON up directional switch at system startup.	• Short circuit of the PCON up directional switch. • GCON ECM.	PCON platform up function inhibited.
U022: GS30, GS32, GS46 & GS47	U022:SWITCHFAULT PCON DOWN	Short circuit of the PCON down directional switch at system startup.	• Short circuit of the PCON down directional switch. • GCON ECM.	PCON platform down functions inhibited.
U023: GS-3232	U023:SWITCHFAULT PCON O/R ENABLE	Short circuit of the PCON outrigger enable switch at system startup.	• Short circuit of the PCON outrigger enable switch. • GCON ECM.	All outrigger functions inhibited.
U033: GS30, GS32, GS46 & GS47	U033:JSTICKFAULT OUT OF CAL RANGE	PCON drive joystick signal is outside acceptable calibration range at system startup.	• PCON drive joystick is not in neutral position at startup. • PCON joystick. • GCON ECM.	All PCON drive and steer functions inhibited.
U034: GS30, GS32, GS46 & GS47	U034:JSTICKFAULT OUT OF RANGE:HI	Short circuit of the PCON drive joystick signal to battery positive at system startup.	• Short circuit of the PCON drive joystick signal circuit. • PCON joystick. • GCON ECM.	All PCON drive and steer functions inhibited.
U035: GS30, GS32, GS46 & GS47	U035:JSTICKFAULT OUT OF RANGE:LO	Short circuit of the PCON drive joystick signal to battery negative at system startup.	• Short circuit of the PCON drive joystick signal circuit. • PCON joystick. • GCON ECM.	All PCON drive and steer functions inhibited.
U036: GS30, GS32, GS46 & GS47	U036:SWITCHFAULT GCON + PCON:ON	Mis-wiring or short circuit of GCON key switch.	• GCON key switch. • GCON ECM.	All functions inhibited.
U037: GS30, GS32, GS46 & GS47	U037:SWITCHFAULT FOOTSW PRESSED	Foot switch pressed at machine startup.	• Short circuit in the foot switch harness. • Foot switch. • GCON ECM.	All functions inhibited.
U038: GS30, GS32, GS46 & GS47	U038:SWITCHFAULT FOOTSWITCH:BAT+	Mis-wiring or short circuit of foot switch to battery positive.	• Short circuit in the foot switch harness. • Foot switch. • GCON ECM.	All functions inhibited.
U039: GS30, GS32, GS46 & GS47	U039:SWITCHFAULT FOOTSW:OPEN/BAT-	Mis-wiring, open or short circuit of foot switch to battery negative.	• Short circuit in the foot switch harness. • Foot switch. • GCON ECM.	All functions inhibited.
U040: GS30, GS32, GS46 & GS47 (GS-3232 excluded)	U040:SWITCHFAULT FOOTSW:TIMEOUT	PCON deck switch pressed at machine startup.	• Short circuit in the foot switch harness. • Foot switch. • GCON ECM.	All functions inhibited.

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DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
U041: GS30, GS32, GS46 & GS47	U041:SWITCHFAULT GCON DECK ENABLE	GCON deck enable switch stuck closed, or depressed at system startup.	• GCON deck enable switch stuck closed. • GCON deck enable switch depressed at system startup. • GCON ECM.	All functions inhibited.
U042: GS30, GS32, GS46 & GS47	U042:SWITCHFAULT PCON DRIVE MODE	PCON drive mode switch stuck closed, or depressed at system startup.	• PCON drive mode switch stuck closed. • PCON drive mode switch depressed at system startup. • PCON ECM.	Drive functions inhibited.
U043: GS-3232	U043:SWITCHFAULT PCON OR MODE	PCON outrigger mode switch stuck closed, or depressed at system startup.	• PCON outrigger mode switch stuck closed. • PCON outrigger mode switch depressed at system startup. • PCON ECM.	Outrigger extend/retract functions inhibited.
U045: GS30, GS32, GS46 & GS47	U045:SWITCHFAULT PCON FUNCTION EN	PCON function enable switch stuck closed, or depressed at system startup.	• PCON function enable switch stuck closed. • PCON function enable switch depressed at system startup. • PCON ECM.	All drive and steer functions inhibited.
U046: GS30, GS32, GS46 & GS47	U046:SWITCHFAULT PCON LIFT MODE	PCON lift mode switch stuck closed, or depressed at system startup.	• PCON lift mode switch stuck closed. • PCON lift mode switch depressed at system startup. • PCON ECM.	All functions operate except platform up/down.

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F001: GS-3232	F001:SWITCHFAULT UP LIMIT1:BAT+	Short circuit of the up limit #1 switch at system startup.	• Short circuit of the up limit switch circuit. • Up limit #1 switch short circuit. • GCON ECM.	Platform up function inhibited.
F003: GS30, GS32, GS46 & GS47	F003:SWITCHFAULT DOWN LIMIT1:BAT+	Short circuit of the down limit #1 switch at system startup.	• Short circuit of the down limit switch circuit. • Down limit #1 switch short circuit. • GCON ECM.	All functions inhibited except platform down.
F005: GS30, GS32, GS46 & GS47	F005:SWITCHFAULT POTHOLE:BAT+	Short circuit of the pothole limit #1 switch at system startup.	• Short circuit of the pothole switch circuit. • Pothole limit #1 switch short circuit. • GCON ECM.	All functions inhibited except platform down as long as machine is in the elevated position. If machine is in stowed position, all functionality is resumed.
F007: GS30, GS32, GS46 & GS47	F007:SWITCHFAULT CHASSISTILT:BAT+	Short circuit of the chassis digital tilt switch at system startup.	• Short circuit of the chassis digital tilt switch circuit. • Chassis digital tilt switch short circuit. • GCON ECM.	All functions inhibited except platform down as long as machine is in the elevated position. If machine is in stowed position, all functionality is resumed.
F008: GS30, GS32, GS46 & GS47	F008:SENSORFAULT OVL D XDUCER:BAT+	Short circuit of the Platform Overload Transducer circuit to battery positive.	• Short circuit in the transducer circuit. • Faulty pressure transducer. • GCON ECM.	All functions inhibited.
F009: GS30, GS32, GS46 & GS47	F009:SENSORFAULT OVL D XDUCER:BAT-	Short circuit of the Platform Overload Transducer circuit to battery negative.	• Short circuit in the transducer circuit. • Faulty pressure transducer. • GCON ECM.	All functions inhibited.
F010: GS30, GS32, GS46 & GS47	F010:SENSORFAULT PLAT HEIGHT:BAT+	Short circuit of the Platform Height Sensor circuit to battery positive.	• Short circuit in the platform height circuit. • Faulty platform height sensor. • GCON ECM.	All functions inhibited except platform down as long as machine is in the elevated position. If machine is in stowed position, all functionality is resumed.
F011: GS30, GS32, GS46 & GS47	F011:SENSORFAULT PLAT HEIGHT:BAT-	Short circuit of the Platform Height Sensor circuit to battery negative.	• Short circuit in the platform height circuit. • Faulty platform height sensor. • GCON ECM.	All functions inhibited except platform down as long as machine is in the elevated position. If machine is in stowed position, all functionality is resumed.

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F012: GS-3232	F012:SENSORFAULT LEVEL PITCH:BAT+	Short circuit of the Level Pitch Sensor circuit to battery positive.	• Short circuit in the level pitch sensor circuit. • Faulty level sensor. • GCON ECM.	All functions inhibited.
F013: GS-3232	F013:SENSORFAULT LEVEL PITCH:BAT-	Short circuit of the Level Pitch Sensor circuit to battery negative.	• Short circuit in the level pitch sensor circuit. • Faulty level sensor. • GCON ECM.	All functions inhibited.
F014: GS-3232	F014:SENSORFAULT LEVEL ROLL:BAT+	Short circuit of the Level Roll Sensor circuit to battery positive.	• Short circuit in the level roll sensor circuit. • Faulty level sensor. • GCON ECM.	All functions inhibited.
F015: GS-3232	F015:SENSORFAULT LEVEL ROLL:BAT-	Short circuit of the Level Roll Sensor circuit to battery negative.	• Short circuit in the level roll sensor circuit. • Faulty level sensor. • GCON ECM.	All functions inhibited.
F016: GS-3232	F016:SENSORFAULT LF RIGGER:BAT+	Short circuit of the Left Front Outrigger Sensor circuit to battery positive.	• Short circuit in the left front outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Left front outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F017: GS-3232	F017:SENSORFAULT LF RIGGER:BAT-	Short circuit of the Left Front Outrigger Sensor circuit to battery negative.	• Short circuit in the left front outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Left front outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F018: GS-3232	F018:SENSORFAULT RF RIGGER:BAT+	Short circuit of the Right Front Outrigger Sensor circuit to battery positive.	• Short circuit in the right front outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Right front outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F019: GS-3232	F019:SENSORFAULT RF RIGGER:BAT-	Short circuit of the Right Front Outrigger Sensor circuit to battery negative.	• Short circuit in the right front outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Right front outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F020: GS-3232	F020:SENSORFAULT LR RIGGER:BAT+	Short circuit of the Left Rear Outrigger Sensor circuit to battery positive.	• Short circuit in the left rear outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Left rear outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F021: GS-3232	F021:SENSORFAULT LR RIGGER:BAT-	Short circuit of the Left Rear Outrigger Sensor circuit to battery negative.	• Short circuit in the left rear outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Left rear outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.
F022: GS-3232	F022:SENSORFAULT RR RIGGER:BAT+	Short circuit of the Right Rear Outrigger Sensor circuit to battery positive.	• Short circuit in the right rear outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Right rear outrigger inhibited if outrigger extend is activated. Outrigger can still be retracted.

Type "FXXX" Faults

Type "FXXX" Faults, continued

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
F023: GS-3232	F023:SENSORFAULT RR RIGGER:BAT-	Short circuit of the Right Rear Outtrigger Sensor circuit to battery negative.	• Short circuit in the right rear outrigger sensor circuit. • Faulty pressure transducer. • GCON ECM.	Right rear outrigger inhibited if outrigger extend is activated. Outtrigger can still be retracted.
F034: GS30, GS32, GS46 & GS47	F034:SENSORFAULT OVLDT:PLAT HEIGHT	Height Sensor settings are not calibrated correctly.	• Height sensor needs calibration for stowed height and maximum height settings. • GCON or PCON ECM.	All functions inhibited.
F035: GS30, GS32, GS46 & GS47	F035:SENSORFAULT OVLDT RANGE:LOW	Height Sensor setting is lower than calibrated stowed height.	• Height sensor not calibrated for stowed height correctly. • GCON or PCON ECM.	All functions inhibited.
F036: GS30, GS32, GS46 & GS47	F036:SENSORFAULT OVLDT RANGE:HI	Height Sensor setting is higher than calibrated maximum height.	• Height sensor not calibrated for maximum height correctly. • GCON or PCON ECM.	All functions inhibited.
F072: GS-2646AV	F072:SENSORFAULT FUNCTION CUT B+	Function Cutout sensor B+ fault.	• Faulty extension deck limit switch. • Faulty gate proximity switch. • GCON ECM.	All functions inhibited.
F073: GS-2646AV	F073:SENSORFAULT DECK SENSOR B+	Power Deck sensor B+ fault.	• Short circuit in deck sensor input circuit. • GCON ECM.	All functions inhibited.
F129: GS30, GS32, GS46 & GS47	F129:TILT SENSOR FAULT	UTS mismatch fault. Two accelerometers inside UTS mismatch.(universal tilt sensor)	• Faulty UTS.	All functions inhibited.

Type "CXXX" Faults

Type "CXXX" Faults

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
C001: GS30, GS32, GS46 & GS47	C001:GCON ECM FAULT TYPE 1	GCON ECM CRC check error.	- Incorrect software file. - GCON ECM internal failure.	All functions inhibited.
C004: GS30, GS32, GS46 & GS47	C004:GCON ECM FAULT TYPE 4	GCON ECM master switch error.	- Short circuit in the master switch circuit. - GCON ECM.	All functions inhibited.
C005: GS30, GS32, GS46 & GS47	C005:GCON ECM FAULT TYPE 5	GCON ECM safety switch error.	- Short circuit in the safety switch circuit. - GCON ECM.	All functions inhibited.
C006: GS30, GS32, GS46 & GS47	C006:GCON ECM FAULT TYPE 6	GCON input redundancy error.	- Input conditioning circuit failure. - GCON ECM.	All functions inhibited.
C007: GS30, GS32, GS46 & GS47	C007:GCON ECM FAULT TYPE 7	GCON ECM inter-processor communication error.	- Incorrectly programmed device. - Error in loading software on device. - GCON ECM.	All functions inhibited.
C009: GS30; GS32' GS46 & GS47	C009:GCON ECM FAULT FAULT TYPE 9	GCON ECM fault type 9.	Contact Genie support.	All functions inhibited.
C010: GS30, GS32, GS46 & GS47	C010:SECONDARY NOT PROGRAMMED	GCON secondary processor not programmed.	- Incorrectly programmed device. - Error in loading software on device. - GCON ECM.	All functions inhibited.
C021: GS30, GS32, GS46 & GS47	C021:PCON NOT DETECTED	PCON not detected error.	- PCON disconnected. - CAN communication failure. - GCON or PCON ECM.	All functions inhibited.
C023: GS30, GS32, GS46 & GS47	C023:MACHINE MODEL FAULT	Discrepancy between model detected and model programmed.	- Incorrect machine model programmed. - GCON or PCON ECM.	All functions inhibited.
C025: GS30, GS32, GS46 & GS47	C025:SYSTEMFAULT PLAT OVLD:NOCAL	Platform overload system not calibrated.	- Platform overload system not calibrated. - GCON or PCON ECM.	All functions inhibited.

Type "CXXX" Faults

Type "CXXX" Faults, continued

DTC Number	Message on GCON LCD	Description	Possible Causes	Failure Mode
C028: GS30, GS32, GS46 & GS47	C028:SERVICE OVERRIDE MODE ON	Machine is in service override mode.	Machine programmed for use in service override mode.	All functions inhibited except for down function and up function. Platform can be elevated only once with the maximum elevate time of XX seconds. Elevate time XX depends on machine model.
C029: GS-3232	C029:SYSTEMFAULT O/R SENSOR:NOCAL	Outrigger not calibrated.	- Outrigger not calibrated. - GCON ECM.	All functions operate. Will not lift above the down limit switch.
C053: GS30, GS32, GS46 & GS47	C053:PCON-GCOM SOFTWARE MISMATCH	Software revisions do not match between the PCON and GCON.	- Short circuit in the safety switch circuit. - GCON ECM.	All functions inhibited.
C055	C055 DCON ECM Fault Type 01	Hardware failur of drive controller	DCON ECM	Performance reduced or all functions inhibited.
C056	C056 DCON ECM Fault Type 02	Setup initialization failure of the drive controller at system startup	- Drive input active at system startup - Faulty drive joystick - Incorrect software D14 - DCON ECM	Performance reduced or all functions inhibited.
C058	C058 DCON ECM Fault Type 04	Drive controller voltage out of range	- Battery charger connected - Batteries too low - DCON ECM	All functions inhibited.
C059	C059 DCON ECM Fault Type 05	Capacitor charge failure of the drive controller	- Open on B+ or B- to the drive contoller - DCON ECM	All functions inhibited.
C061	C061 DCON ECM Fault Type 07	Open circuit of the key switch circuit or battery positive / negative of the drive controller	- Open circuit of the motor controller harness - Open circuit to the B+ and/or B- - Key switch relay power not closed - DCON or GCON ECM	All functions inhibited.
C063	C063 DCON ECM Fault Type 09	Communication error of the CAN circuit between the GCON and the motor controller	- Open circuit of the motor controller harness - DCON or GCON ECM	All functions inhibited.
C064	C064 DCON ECM Fault Type 10	Output error of the thermal sensor circuit of the motor controller	DCON ECM	All functions inhibited.

Note: An error code C053 will be displayed if an updated PCON is installed on a machine that has not had the GCON updated. The GCON must be updated, or the original PCON must be installed back on the machine.

Battery Charger

Battery Charger

The charger continuously monitors internal and external conditions. Fault and Error codes are generated by the charger to indicate that an internal or external malfunction has been detected in the electrical system. The types of Diagnostic Trouble Codes that may occur are explained below.

Type "F" codes – Indicate an internal fault condition has occurred and caused charging to stop.

Type "E" codes - Indicate an external fault condition has occurred and caused charging to stop.

Type "P" code - Indicates charger programming mode is active. This will occur when the charger profile is being configured.

Type "USB" code - Indicates the USB interface is active, and the USB should not be removed. This will occur when the charger firmware is being updated.



Solid red - Internal fault condition. Type "F" fault code.



Flashing amber - External fault condition. Type "E" error code.



Flashing green - USB interface is active.

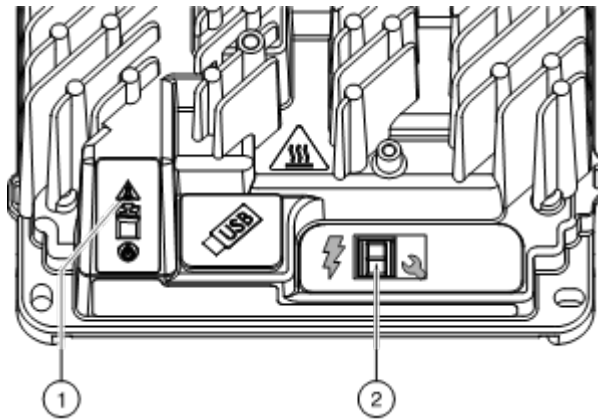


Solid green - Firmware update is complete. Remove USB from charger.

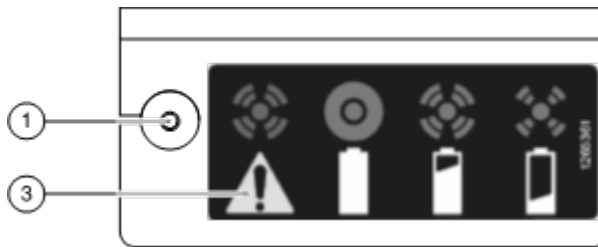
Codes are displayed as F, E or P and followed by three numbers and a period (example: E-0-0-4). The charger can display multiple codes.

If the batteries are changed from one type to another, (example: flooded lead acid to AGM), the charger profile must be programmed. Refer to Repair Procedure, *Programming the Charger Profile*.

Battery Charger



Charger status indicators



Remote charger status indicators

- 1 Error / Fault / USB indicator
- 2 Charger profile / Error display
- 3 Error / Fault

Charger Fault Codes

Charger Fault Code	Genie Fault ID	Genie Fault Type	Description	Possible Causes	Solution
F-0-0-1	135	8	DC-DC failure: LLC excessive leakage fault.	Internal charger fault.	Disconnect AC input and battery pack for 30 seconds. If fault does not clear, replace charger.
F-0-0-2	135	8	PFC failure: PFC excessive leakage fault.		
F-0-0-3	135	8	PFC taken too long to boost.		
F-0-0-4	135	8	Charger unable to calibrate current offset.		
F-0-0-5	135	8	Output relay voltage too high when closed.		

Battery Charger

Charger Error Codes

Charger Error Code	Genie Fault ID	Genie Fault Type	Description	Condition	Possible Causes
E-0-0-1	131	23	High battery voltage	Battery voltage too high to charge.	Check battery voltage and cable connections. Self clearing when condition has been corrected.
E-0-0-2	131	24	Low battery voltage detected prior to starting charge cycle	Battery voltage too low to charge.	Check battery voltage and cable connections. Check battery condition. Self clearing when condition has been corrected.
E-0-0-3	135	28	Charge timeout	Charge timeout caused by battery pack not reaching required voltage within safe time limit.	Charge output reduced due to high temperature. Charge at lower temperature. Check battery condition. Replace damaged battery. Check battery connections. Battery pack deep discharge. Disconnect battery pack for 30 seconds to clear fault when condition has been corrected.
E-0-0-4	204	21	Defective battery	Battery pack could not be trickle charged up to the minimum voltage.	Check battery condition. Replace damaged battery. Check battery connections. Disconnect battery pack for 30 seconds to clear fault when condition has been corrected.
E-0-0-7	264	23	Amp hour limit exceeded	Safety limit exceeded.	Check battery condition. Replace damaged battery. Check battery connections. Battery pack deep discharge. High parasitic loads on battery pack while charging. Disconnect parasitic loads. Disconnect battery pack for 30 seconds to clear fault when condition has been corrected.

Battery Charger

Charger Error Codes

Charger Error Code	Genie Fault ID	Genie Fault Type	Description	Condition	Possible Causes
E-0-0-8	233	19	Battery temperature out of range	Battery temperature sensor.	Check temperature sensor and electrical connections. Rest charger. Disconnect AC power for 30 seconds. Self clearing when condition has been corrected.
E-0-1-1	135	8	Charger disabled by external command	Battery pack not being charged..	Power outage. AC power disconnected before charging complete.
E-0-1-2	135	8	Reverse polarity	Battery connected incorrectly.	Check battery connections. Self clearing when condition has been corrected.
E-0-1-3	204	21	Battery does not take current	Battery voltage is detected, but charger is unable to charge the battery pack..	Incorrect profile for battery type. Installed from a different machine with different battery type. Disconnect battery pack for 30 seconds to clear fault when condition has been corrected.
E-0-1-6	135	8	Software update failed	Software update failed.	Confirm USB Flash Drive is formatted with correct software. If software update continues to fail, contact Genie Product support.
E-0-1-7	265	21	USB error	USB not connected correctly.	Remove and re-insert USB Flash Drive Disconnect AC input and battery pack for 30 seconds.
E-0-1-8	135	8	Slot CRC error	Software update failed.	Confirm USB Flash Drive is formatted with correct software. If software update continues to fail, contact Genie Product support.
E-0-1-9	135	29	Hardware does not support software	Charger hardware does not support software version being programmed.	Confirm USB Flash Drive is formatted with correct software. If software update continues to fail, contact Genie Product support.
E-0-2-0	266	31	No algorithm selected	Charger profile not programmed.	Program the charger for the proper battery profile.

Battery Charger

Charger Error Codes

Charger Error Code	Genie Fault ID	Genie Fault Type	Description	Condition	Possible Causes
E-0-2-1	254	23	High battery voltage detected while charging	Battery voltage too high as detected by charger profile.	Check battery voltage and cable connections. Confirm charger profile is programmed for correct battery type. Self clearing when condition has been corrected.
E-0-2-2	254	24	Low battery voltage detected while charging	Battery voltage too low as detected by charger profile.	Check battery voltage and cable connections. Confirm charger profile is programmed for correct battery type. Self clearing when condition has been corrected.
E-0-2-3	267	23	High AC voltage	AC voltage greater than 270 VAC.	Connect to a stable AC source between 85 - 270 VAC / 45 - 65 Hz. Self clearing when condition has been corrected.
E-0-2-4	135	8	Failure to initialize	Charger failed to turn on correctly.	Disconnect AC input and battery pack for 30 seconds. Contact Genie Product Support.
E-0-2-5	267	24	Low AC voltage oscillation	AC voltage unstable.	Connect to a stable AC source between 85 - 270 VAC / 45 - 65 Hz. Using an undersized generator. Self clearing when condition has been corrected.
E-0-2-6	135	8	Script failure	Software update failed.	Confirm USB Flash Drive is formatted with correct software. If software update continues to fail, contact Genie Product support.
E-0-2-7	265	21	USB over current fault	USB over current protection has been tripped.	Remove and re-insert USB Flash Drive Use different USB Flash drive.
E-0-2-8	266	21	Charge profile incompatibility	Selected charger profile is not compatible with charger software.	Update charger software. Confirm charger profile is programmed for correctly. Updating current firmware and error occurs, contact Genie Product Support.

Battery Charger

Charger Error Codes

Charger Error Code	Genie Fault ID	Genie Fault Type	Description	Condition	Possible Causes
E-0-2-9	135	18	CAN Bus error	CAN Bus network error.	Check CAN Bus connections. CAN modules not functioning correctly. Termination resistance is ~60 ohms.
E-0-3-0	135	8	COMM battery module error	CAN Bus battery module error.	CAN Buss battery module not functioning correctly.
E-0-3-1	135	8	Vref for ADC measurements triggered alarm	Internal charger error.	Disconnect AC input and battery pack for 30 seconds. Contact Genie Product Support.
E-0-3-2	135	13	CAN Bus heartbeat error	CAN Bus heartbeat error.	CAN Bus devices not functioning correctly.
E-0-3-6	233	21	Battery temperature sensor charge profile	Battery temperature sensor is required by charger profile but is not installed.	Check battery temperature sensor connections. Check charger profile and confirm battery temperature sensor is installed.
E-0-3-7	135	8	CAN Open Error	CAN Open reprogramming failed.	Download CAN Open software. Reprogram charger using USB Flash Drive.
E-0-3-8	268	21	Stalled cooling fan	Fan obstructed.	Inspect fan for damage. Clear blockage.